RL Environments for Finance

A technical report on reinforcement learning environments for financial decision-making.

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The Research Engine

The research engine is the synthesis of decades of financial experience: the judgment of practitioners who have priced risk through cycles, combined with an immense corpus of material gathered from the broadest possible range of sources, including regulatory filings, transcripts, market microstructure, credit documentation, and primary research. It is not a single model but an expression of those materials, their knowledge, and the techniques used to interrogate them.

Operationally, the engine is a **multi-agent system**. A central orchestrator decomposes a question, whether about a company, a sector, or a specific security, into focused tasks and dispatches them to specialised agents for fundamentals, valuation, credit, and market context. Each agent draws on dedicated data stores and a unified knowledge graph that links entities, instruments, and events, so that a claim about a stock is grounded in the same fabric as a claim about its debt. Agents critique and refine one another's output, and the system loops recursively until the analysis converges.

Crucially, the loop is closed by people. A human interface lets analysts steer the inquiry, challenge conclusions, and inject judgment that the corpus alone cannot supply, and every interaction feeds back into the engine, sharpening it over time. The result is a research capability that spans the full capital structure, from equities to debt, with the depth of an institution and the speed of software.

FIGURE 1 · ARCHITECTURE OF THE MULTI-AGENT RESEARCH ENGINE